

## Introduction to Statistical Methods

### (S2-25 AIMLCLZC418) – Assignment 1

#### AIML – Section 2

**Question 1&2 carries 2.5 marks each.**

**Question 3 carries 5 marks**

**Total - 10 Marks**

**Duration: 30<sup>th</sup> May 2026 – 14<sup>th</sup> June 2026**

- 1) Submissions are individual**
- 2) Solve these on paper, scan, and upload**
- 3) Plagiarism results in zero marks**
- 4) Write your name, BITS ID and Section on each page**
- 5) Only handwritten solutions with formula, full steps with proper justification are required.**

Answer all the questions

1. In a survey it is found that 50% of the men like Tea, 40% like Coffee, 30% like Juice, 20% like Tea & Coffee, 15% like Coffee & Juice, 10% like Tea & Juice, 5% like all three. Find the probability that a person likes at least one drink. **[2.5 M]**
2. The probability that a man hits the target is  $\frac{1}{3}$ . If he fires 6 times what is the chance of hitting i) Exactly once ii) at least once iii) at most 5 times. **[2.5 M]**
3. A university offers scholarships. 50% of applicants are from urban areas. 30% are from rural areas. 20% are international students. A randomly selected applicant received a scholarship.
  - a) Find the probability that the student is from a rural area.
  - b) Find the probability that the student is from an urban area.
  - c) Which category benefits most?
  - d) Interpret the fairness of the process.
  - e) If a scholarship recipient is chosen at random, what is the probability that the student is either rural or international? **[5 M]**

----- ALL THE BEST-----